

INFLURIOS

INFLR.me Short-Link & Dynamic QR Development Specification

Version 1.0 • Build-ready specification for Development AI

Platform	Influrios
Short-link domain	INFLR.me
Canonical profile host	influrios.com
Architecture position	Same Influrios infrastructure; separate logical short-link service

Non-negotiable: INFLR.me is infrastructure for compact identity links and permanent QR resolution. It is not a second consumer brand and it is not the canonical content host.

1. Purpose and Product Role

INFLR.me is the compact identity and redirect layer for Influrios. It powers shareable creator links, QR codes, future NFC cards, event badges, email signatures, business cards, and other creator identity surfaces while the canonical creator profile remains on Influrios.

- Canonical creator profile example: <https://influrios.com/@sofia>
- Short creator identity link example: <https://inflr.me/sofia>
- Permanent QR resolver example: <https://inflr.me/q/K7xP92m>
- INFLR.me must remain visually and operationally subordinate to the Influrios brand.
- The system must be provider- and domain-configurable so a fallback short domain can be introduced later without code changes.

2. Required Architecture

Use the same Influrios cloud/backend ecosystem, database environment, admin portal, logging and security controls. Implement INFLR.me as a separate logical service/module optimized for very fast redirect resolution. For MVP, it may share the same backend deployment and primary database if that is operationally simpler.

Preferred request flow

INFLR.me → CDN/Edge → Short-Link Resolver → Cache → Short-Link Store → Analytics Queue → Redirect Destination

- Do not require the full Influrios application to boot before a redirect is resolved.
- Keep the resolver stateless where practical.
- Cache active redirect records aggressively with safe invalidation after admin or creator changes.
- Target redirect latency below 200 ms in normal operating regions.
- The main Influrios web app and the short-link resolver must be independently deployable later.

3. Domain and Routing Configuration

Route	Purpose	Notes
/<slug>	Public creator short link	e.g. inflr.me/sofia
/q/<token>	Permanent QR identity	Never expose creator DB id; token must survive slug changes
/c/<token>	Campaign links	Future/optional
/i/<token>	Invitation links	Future/optional
/r/<token>	Referral links	Future/optional

Admin must manage short-link domains under Branding / Short-Link Domains. INFLR.me is the primary production domain. A fallback such as links.influrios.com may be configured for staging, recovery or future migrations.

4. Core Data Model

Entity	Required fields / role
short_link	id, creator_id, domain_id, slug, destination_type, destination_url, status, is_dynamic, entitlement_snapshot, created_at, updated_at
short_link_alias	id, short_link_id, old_slug, active_redirect, created_at
qr_identity	id, creator_id, short_link_id, token, status, created_at, rotated_at
short_link_domain	id, hostname, environment, verified, active, is_primary, created_at
short_link_event	id, short_link_id/qr_identity_id, event_type, timestamp, coarse_geo, device_class, referrer, campaign_id, privacy_metadata
reserved_slug	id, slug, reason, scope, created_by, created_at
short_link_abuse_case	id, short_link_id, report_type, status, reviewer, actions, timestamps

5. Slug and Handle Rules

- A creator's database identifier must never depend on the public slug.
- Slugs are aliases only and may change.
- Changing a slug must not break old published links unless explicitly disabled for abuse/security reasons.
- Old active slugs should redirect to the creator's current short link by default.
- Admin can reserve, block, release, transfer or suspend a slug.
- Prevent impersonation, protected operational words, obvious fraud terms and unsafe paths.
- Reserve internal words such as admin, login, signup, support, help, billing, api, www, mail, privacy, terms, influrios and inflr.
- Creator-facing slug availability checks must be real-time and race-safe.

6. Subscription and Entitlement Integration

Do not check plan names directly. All short-link behavior is driven by the central Influrios entitlement engine.

Capability	Starter default	Plus default	Pro default
Canonical Influrios profile	Yes	Yes	Yes
INFLR.me short link	No	Yes	Yes
Custom slug	No	Yes	Yes
Standard QR	No	Yes	Yes
Dynamic QR	No	No	Yes
QR/short-link analytics	Basic profile only	Standard	Advanced
Custom redirect destination	No	No	Yes
Multiple short links	0	Admin-configurable	Admin-configurable

Initial defaults above are commercial configuration only. Admin must be able to change them without code changes.

7. QR Identity Design

The QR code must resolve through a permanent opaque QR token, not directly to the creator's current slug or canonical profile URL. This is essential so printed cards, signage, packaging, NFC companion assets and event materials continue to work after a creator changes handle or destination.

- Visible link may be `inflr.me/sofia`.
- QR payload should be an opaque route such as `inflr.me/q/K7xP92m`.
- The QR identity maps to the active short-link record.
- Slug changes do not require QR regeneration.
- For Pro dynamic QR, destination changes do not require QR regeneration.
- Token rotation is an exceptional security action and must be audited.
- QR generation must produce machine-readable SVG and PNG variants, but live destination information must not be baked into the visual beyond the encoded token.

8. Redirect Resolution Logic

1. Receive host + route and normalize safely.
2. Validate route namespace and rate limits.
3. Resolve from edge/cache where possible.
4. If cache miss, retrieve the active short-link or QR identity record.
5. Check status: ACTIVE, SUSPENDED, EXPIRED, DISABLED, REDIRECTED or NOT_FOUND.
6. Check entitlement only where the product requires ongoing entitlement enforcement; do not break previously issued links without an explicit downgrade policy.
7. Determine destination according to static/dynamic rules.
8. Record privacy-safe analytics asynchronously.
9. Issue an HTTP redirect to the destination.
10. Return a branded fallback page for unavailable/suspended links rather than a raw server error.

Use appropriate redirect status codes based on product behavior. The Dev AI should document when 301, 302, 307 or 308 is used and avoid caching mistakes that make dynamic destination changes impossible.

9. Dynamic Redirect Rules for Pro

- Default destination should remain the Influrios Card/Profile.
- Allow only destination classes explicitly enabled by Admin.
- External destinations must pass safety checks.
- Store full destination history and the actor who changed it.

- Allow immediate rollback to the prior safe destination.
- Future campaign routing may support scheduled start/end dates and campaign tags.
- Never allow Pro dynamic routing to bypass account suspension, legal takedown, abuse restrictions or jurisdiction controls.

10. Analytics

Analytics must be useful without turning the short-link layer into invasive tracking infrastructure.

- Track total visits, QR scans, direct short-link visits, referrer class, device class, coarse geography where permitted, social/contact CTA clicks and business/collaboration inquiry conversions.
- Avoid storing unnecessary precise location.
- Apply privacy/consent rules by region.
- Support bot filtering and internal-traffic exclusion.
- Creator analytics level is entitlement-driven.
- Admin analytics can aggregate traffic, abuse, resolver performance and top route health.
- Never expose another creator's detailed traffic to other creators.

11. Open Graph and Social Sharing

- Shared INFLR links should generate an Influrios-branded preview using the creator's approved profile image, name and specialty summary.
- The final canonical content and SEO ownership remain on influrios.com.
- INFLR.me should generally redirect rather than host duplicate profile pages.
- If a preview interstitial is required for messaging-platform previews, it must set canonical references back to Influrios and must not become a duplicate SEO destination.

12. Admin Portal Requirements

- Dashboard of total links, QR identities, scans, errors and abuse cases
- Search creators/links by slug, creator id, status or domain
- Create/disable/reactivate short links
- Reserve or release slugs
- Transfer slug after approved identity/ownership review
- View alias/redirect history
- View QR association and token status
- Change allowed destination subject to audit
- View analytics appropriate to staff role
- Manage short-link domains and verification status

- Manage blocked destination domains and route rules
- Manage reserved/prohibited slug vocabulary
- Review abuse reports and suspend links
- View complete audit trail

13. Creator Experience

- Show creator's canonical Influrios URL and INFLR short URL together.
- Allow eligible creators to search/select a custom slug.
- Provide clear copy/share actions.
- Show QR preview and downloadable versions when entitled.
- Explain if an action is locked by entitlement and link to the relevant upgrade path.
- Warn before a slug change and explain old-link behavior.
- For Pro, provide destination management and advanced analytics without exposing unsafe low-level redirect controls.

14. Security and Abuse Controls

- Rate-limit route lookups, availability checks and admin mutations.
- Detect or block phishing, malware, deceptive impersonation and prohibited external destinations.
- Require authorization/verification for custom external destinations.
- Maintain admin suspension and emergency disable controls.
- Log all privileged changes.
- Support user and public abuse reporting.
- Prevent open-redirect vulnerabilities.
- Normalize and validate Unicode/IDN/slug inputs to reduce spoofing.
- Protect analytics endpoints from enumeration.
- Do not leak internal creator ids in public routes.

15. Reliability and Fallback Behavior

- A creator short link should continue resolving even if the main Influrios frontend is temporarily unavailable, where the destination itself remains available.
- Use cache and fallback configuration appropriate to the deployment.
- Suspended creator: show a branded unavailable page or route to a controlled unavailable endpoint.
- Unknown slug: show a minimal branded 404 with a path back to Influrios.
- Resolver failure must not leak stack traces or infrastructure details.

- Monitor redirect success rate, p50/p95 latency, error rate, cache hit rate and webhook/analytics queue health.

16. API Surface

Endpoint	Purpose
POST /api/short-links	Create eligible short link
GET /api/short-links/{id}	Read owner/admin short-link detail
PATCH /api/short-links/{id}	Update allowed fields/destination
POST /api/short-links/check-slug	Check availability
POST /api/short-links/{id}/change-slug	Perform race-safe slug change
POST /api/qr-identities	Create permanent QR identity
GET /api/short-links/{id}/analytics	Entitlement-aware analytics
POST /api/admin/short-links/{id}/suspend	Admin suspension
POST /api/admin/slugs/reserve	Reserve slug
POST /api/admin/short-domains	Add/verify domain

17. Events and Audit

- short_link_created
- short_link_slug_changed
- short_link_destination_changed
- short_link_suspended
- short_link_reactivated
- qr_identity_created
- qr_identity_rotated
- short_link_visit
- qr_scan
- short_link_cta_click
- short_link_abuse_reported

Business analytics events and immutable security/admin audit events are separate concerns and should not be conflated.

18. Development AI Guardrails

- **Do not build INFLR.me as an unrelated application with a separate creator/account system.**
- **Do not host duplicate canonical creator profiles on INFLR.me.**

- Do not hard-code INFLR.me into all application logic; read active short-link domain configuration.
- Do not encode canonical profile URLs directly into permanent QR codes.
- Do not use public slugs as primary keys.
- Do not tie features to literal plan names.
- Do not implement an unrestricted open redirect.
- Do not silently break existing QR codes when slugs change.
- Do not store unnecessary precise geolocation for analytics.
- Do not make short-link redirects dependent on full frontend page rendering.
- Do not expose internal identifiers, secret provider data or admin-only metadata.

19. Phased Implementation

Phase	Scope	Launch dependency
Phase 1	Domain verification, root slug resolver, short_link model, admin management, Plus short links	Yes
Phase 2	Permanent QR identity, standard QR generation, alias history, basic analytics	Yes for QR-enabled tiers
Phase 3	Advanced analytics, Pro dynamic destination, richer abuse controls	Post-core / Pro
Phase 4	Campaign links, NFC workflows, scheduled routing, enterprise/agency extensions	Future

20. Definition of Done / Acceptance Criteria

11. INFLR.me/validslug resolves to the correct current Influrios destination.
12. Changing a creator slug preserves old-link resolution according to configured alias policy.
13. A permanent QR token remains valid after a slug change.
14. A Pro dynamic destination change does not require QR regeneration.
15. A suspended short link cannot redirect to the previous destination.
16. Starter/Plus/Pro behavior follows entitlements, not literal plan names.
17. Admin can reserve, suspend and audit slugs/links without code changes.
18. Unknown or disabled links fail safely with branded UX.
19. Redirects are protected from open-redirect abuse.
20. Analytics respect configured privacy rules and entitlements.
21. Main Influrios profile remains the canonical SEO destination.

- 22. Resolver performance and errors are observable in production.
- 23. All privileged changes are auditable.
- 24. Production configuration can change the primary short domain without code changes.

21. Canonical Developer Instruction

INFLR.me is a redirect and identity infrastructure service inside the Influrios ecosystem. Creator profiles remain canonical on Influrios. All INFLR links must resolve through database-backed, auditable short-link records. Permanent QR identities must remain valid when slugs, profile routes or destinations change. Short-link features are entitlement-driven and admin-configurable. The service may share the same backend/server environment at MVP, but its resolver must remain logically separable and optimized for fast independent redirect handling.

End of Specification • Influrios / INFLR.me v1.0